



PRODUCT  
DATASHEET '25-26'

# ADVANCE ANALYTIK

REVOLUTIONIZING ONLINE MONITORING SOLUTIONS



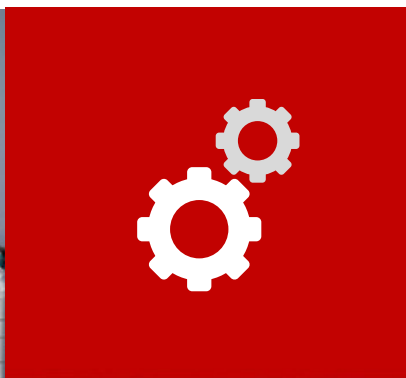
## Optics-1000 Boron (Lr)

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# Optics-1000 Boron (Lr)

## Method - Colorimetric

After adding the sample into the measurement cell, some reagents are added in order to adjust the solution to the desired conditions (pH, valence's elements, etc.). Then, a blank is done to correct any temperature or turbidity disturbance. Subsequently, a last reagent is added, and it reacts with solution developing a color, which is measured using a correct wavelength. Thanks to the photometer used, the result achieves a great accuracy.



## Principle of measurement

Thanks the buffer reagent, the sample is adjusted to pH between 5 and 6. Then, the addition of *Azomethine-H* & *Ascorbic acid*, they react with Boron giving a green-yellow color that is measured at 420nm.

## Advantages of the method

The method is accurate and sensitive. The range is linear up to 2 ppm and higher ranges can be obtained diluting the sample.



# Optics-1000 Boron (Lr)

## Technical Specifications

| Basic Specifications |                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| Range                | From 0 to 100 ppb / 500 ppb / 1000 ppb / 2000 ppb.<br>(Adjustable higher concentrations adjusting reagents.) |
| Accuracy             | ±2%                                                                                                          |
| Repeatability        | ±2%                                                                                                          |
| Resolution           | 0,1 ppb                                                                                                      |
| Analysis time        | around 30 minutes                                                                                            |
| Calibration          | Two point                                                                                                    |
| LED Wavelength       | 420 nm                                                                                                       |
| Reagents consumption |                                                                                                              |
| Reagent 1            | 2.2 ml / analysis - 2.0L / month                                                                             |
| Reagent 2            | 4.2 ml / analysis - 3.5L / month<br>(Monthly consumption calculated assuming 1 analysis per hour.)           |





# APPLICATIONS

- Semiconductor industry
- Desalination plants and more.



**Aquaculture Farms**  
Maintain Hd levels in fish tanks for optimal aquatic conditions.



**Drinking Water Quality**  
Ensure safe drinking water by monitoring Hd levels in water sources.



**Environmental Research**  
Assess Hd concentrations in water bodies to study pollution and ecosystem health.



**Agriculture and Irrigation**  
Monitor Hd levels in irrigation water to optimize nutrient supply to crops.



**Scientific Laboratories**  
Conduct Hd analysis for research and educational purposes.



**Industrial Processes**  
Control Hd concentrations in wastewater to meet environmental regulations.



## Note -

*This data sheet serves as general information about the Optics-1000 Boron (Lr). For specific technical details, installation guidelines, and troubleshooting assistance, please refer to the official user manual provided with the product.*

*For inquiries and detailed technical information, please contact [sales@advanceanalytik.com](mailto:sales@advanceanalytik.com).*



# LET'S CONNECT !

Let's work together to find a solution that works for you



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